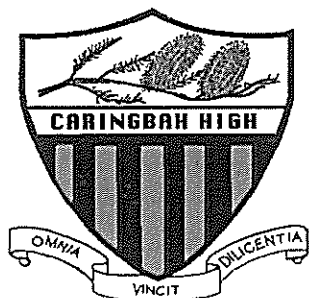


Caringbah High School

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Student Name



2013 Year 11
Semester 1 Examination

Mathematics (2 Unit)

General Instructions

- Approved calculators permitted.
- Start each question in a new booklet.
- Answers without working, mathematical reasoning and/or calculation may not attract full marks.
- Marks may not be awarded for carelessly arranged work.

Time Allowed

2 hours + 5 minutes reading time

Question 1 (12 marks) Start a NEW booklet.	Marks
a) Calculate $\sqrt[3]{1.35 \times 0.479}$ to 3 significant figures	1
b) Change 68.69° to degrees, minutes and seconds	1
c) Find x if $\tan 20^\circ = \cot (x + 30)^\circ$	2
d) Express as a simplified surd:	
(i) $\sqrt{192}$	1
(ii) $\sqrt{147} - 4\sqrt{3}$	1
(iii) $4\sqrt{3} \times \sqrt{48}$	1
(iv) $\frac{\sqrt{125}}{5}$	1
e) If $\sin \theta = 0.45$ and θ is acute, find θ to the nearest minute	1
f) Change $N40^\circ W$ to a 3 digit bearing	1
g) If $a = 4$, $b = -2$ and $c = -3$, find the value of $3(bc)^2 - a + c$	2

Question 2 (12 marks) Start a NEW booklet.	Marks
a) Simplify, in exact form, the value of $\cos 30^\circ + \sin 45^\circ$, with a rational denominator	2
b) Factorise $x^2 + 5x - 6$	1
c) Solve $4x + 7 = 3(x - 2)$	2

Question 2 continued over/

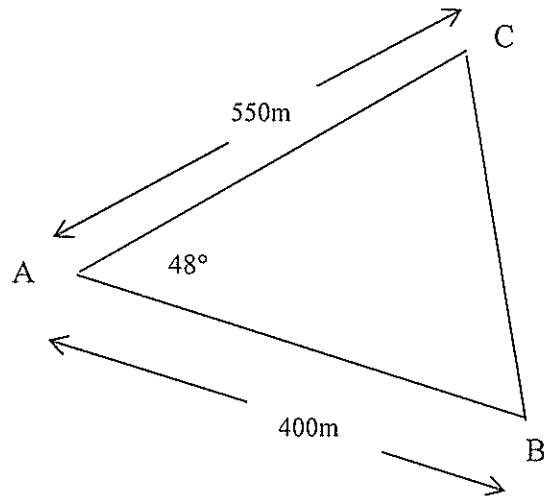
d) Solve:

(i) $3^{x-1} = 9^{2x}$ 2

(ii) $5 - 3x < 7$ 1

(iii) $|2x + 5| < 3$ 2

e) In $\triangle ABC$, $AC = 550m$, $AB = 400m$ and $\angle CAB = 48^\circ$. Calculate the length of BC , to the nearest metre. 2



NOT TO SCALE

Question 3 (12 marks) Start a NEW booklet.

Marks

a) Express $0.\dot{2}\dot{3}$ as a fraction 2

b) If $\tan \theta = -\frac{2}{3}$ and $\sin \theta > 0$, find the exact value of:

(i) $\cot \theta$ 1

(ii) $\sin \theta$ 2

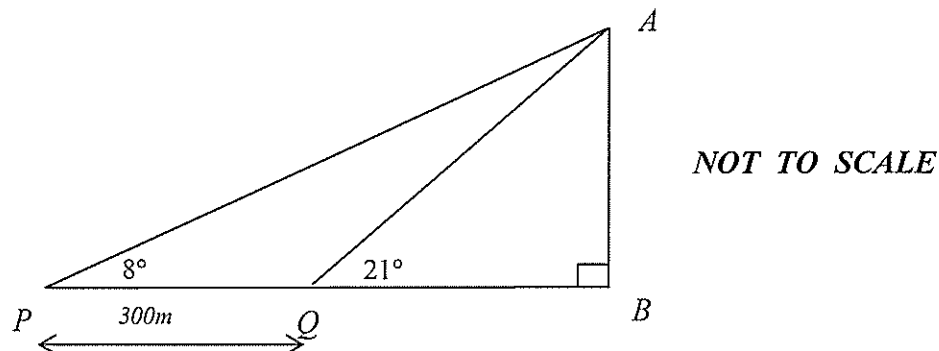
c) Solve $|x - 1| = 4$ 2

d) Solve $\sin \theta + \sqrt{3} \cos \theta = 0$ for θ , where $0^\circ \leq \theta \leq 360^\circ$ 3

e) Simplify $1 + \tan^2(90 - \theta)$ 2

Question 4 (12 marks) Start a NEW booklet.**Marks**

- a) From point P , Marc finds that the angle of elevation of the top A of a rock pillar AB is 8° . After walking $300m$ directly towards the pillar to the point Q he finds that the angle of elevation of A is 21° .

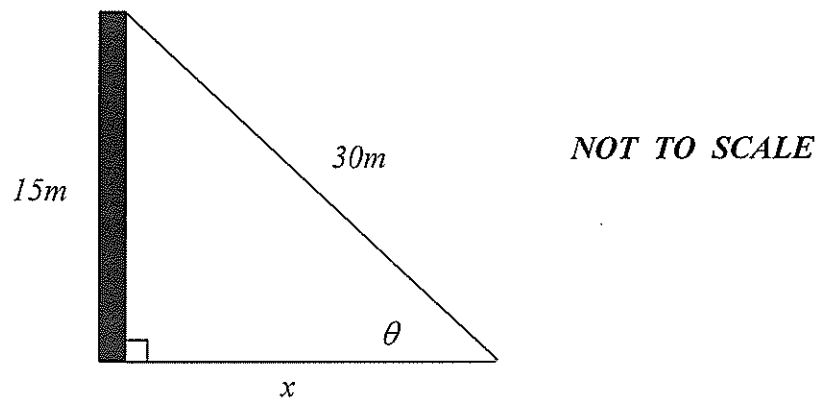


- | | | |
|-------|---|---|
| (i) | Copy the diagram and find $\angle PAQ$. | 1 |
| (ii) | Calculate the length of AQ . | 2 |
| (iii) | Find the height of the rock pillar AB . | 2 |
- b) Simplify:
- | | | |
|-------|---|---|
| (i) | $\frac{2x^2 - 3xy}{xy - y^2} \div \frac{4x - 6y}{2x^2 - 2xy}$ | 2 |
| (ii) | $\frac{a^3 + b^3}{a^2 - b^2}$ | 2 |
| (iii) | $\frac{2}{m^2 - 4} - \frac{1}{m^2 - 3m + 2}$ | 3 |

Question 5 (12 marks) Start a NEW booklet.

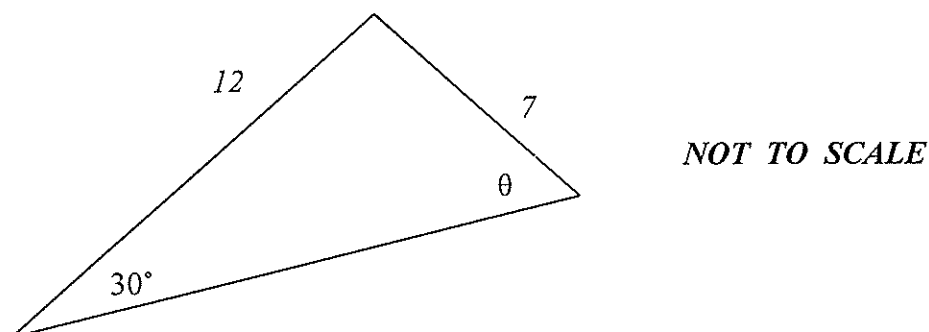
Marks

- a) A vertical pole of height $15m$ stands on the level ground and a straight wire $30m$ long joins the top of the pole onto the ground.



Find:

- | | | |
|-----------------------------|--|---|
| (i) | the distance x of this point from the foot of the pole (in exact form) | 1 |
| (ii) | the angle the wire makes with the ground | 2 |
| b) Find the exact value of: | | |
| (i) | $\sin 240^\circ$ | 1 |
| (ii) | $\cos(30)^\circ$ | 1 |
| (iii) | $\operatorname{cosec} 45^\circ$ | 1 |
| (iv) | $\cot 420^\circ$ | 1 |
| c) | Find all possible value/s of θ , to the nearest degree. | |



- | | | |
|----|--|---|
| d) | Express $\frac{\sqrt{2}-1}{2\sqrt{2}-1}$ in the form $a+b\sqrt{2}$ and state the values of a and b . | 2 |
|----|--|---|

Question 6 (12 marks) Start a NEW booklet.

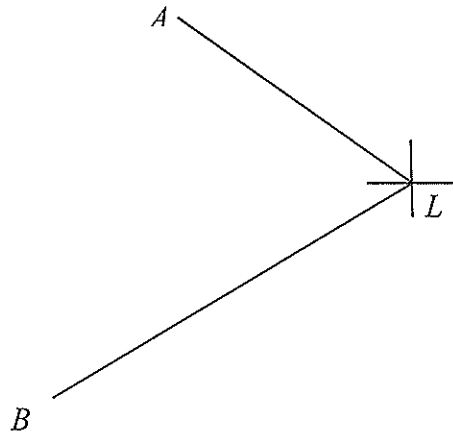
Marks

a) Simplify $\sqrt{169a^6b^2}$

1

- b) A is 5km from a lighthouse on a bearing of 323° . Also, B is 12km and at a bearing of 233° from the same lighthouse.

- (i) Copy and complete the diagram showing the above information on your page, $\frac{1}{3}$ page size.



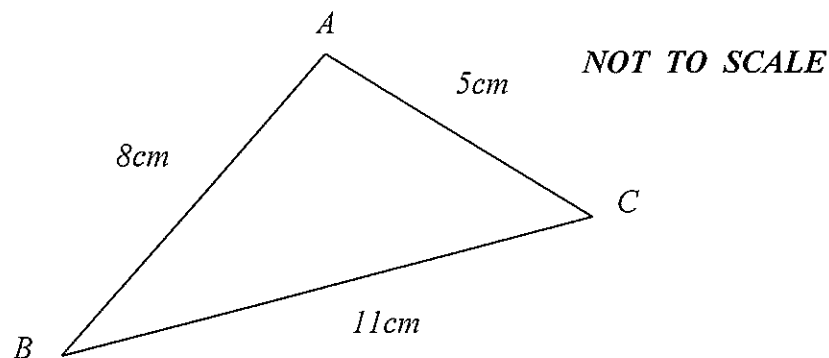
- (ii) Calculate the distance of AB .

1

- (iii) Find the bearing of B from A .

2

- c) In $\triangle ABC$, $BC = 11\text{cm}$, $AC = 5\text{cm}$ and $AB = 8\text{cm}$.



- (i) Calculate $\angle ABC$, to the nearest degree.

2

- (ii) Find the area of $\triangle ABC$

2

- d) Solve $2x^2 = x + 5$ by using the quadratic formula, leaving your answer in exact surd form.

2

- e) Sketch the graph $y = \cos \theta$ for $0^\circ \leq \theta \leq 360^\circ$, using a $\frac{1}{3}$ page size.

2

Question 7 (12 marks) Start a NEW booklet.

Marks

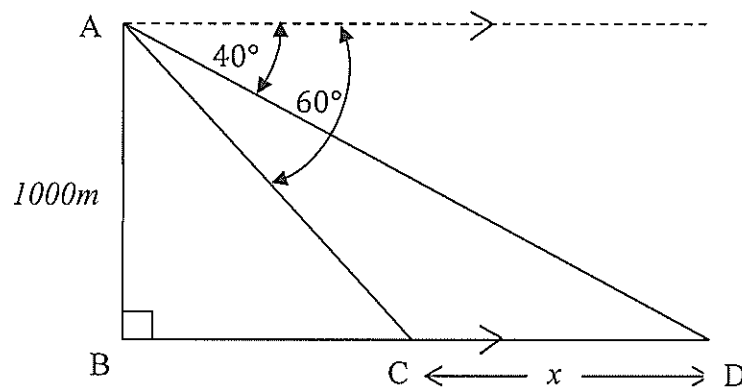
- a) Solve $x^2 - 10x - 1 = 0$ by completing the square, leaving your answer in exact form.

2

- b) Solve simultaneously $\begin{cases} 3x - 2y = 14 & \text{---[1]} \\ xy = 12 & \text{---[2]} \end{cases}$

3

- c) From an aircraft 1000m above the ground, the angle of depression of the top of two houses (ignoring their height) in line with the aeroplane are 40° and 60° respectively. How far apart are the houses?



3

- d) Prove $(1 - \tan x)^2 + (1 + \tan x)^2 = 2 \sec^2 x$

2

- e) If $x = 5 \sin \theta$ and $y = 5 \cos \theta - 1$, eliminate θ to find an expression relating x and y .

2

End of Examination

Q1
(a) $\sqrt[3]{1.35 \times 0.479} = 0.865$

(b) $68-69^\circ = 68^\circ 41' 24''$

(c) $\tan 20^\circ = \cot (x+30^\circ)^\circ$
 $\tan 20^\circ = \cot (90-20)^\circ$
 $= \cot 70^\circ$
 $\therefore x+30=70$
 $x=40^\circ$

(d) (i) $\sqrt{192} = \sqrt{64 \times 3}$
 $= 8\sqrt{3}$

(ii) $\sqrt{147} - 4\sqrt{3} = \sqrt{49 \times 3} - 4\sqrt{3}$
 $= 7\sqrt{3} - 4\sqrt{3}$
 $= 3\sqrt{3}$

(iii) $4\sqrt{3} \times \sqrt{48} = 4\sqrt{3} \times \sqrt{16 \times 3}$
 $= 4\sqrt{3} \times 4\sqrt{3}$
 $= 48$

(iv) $\frac{\sqrt{125}}{5} = \frac{5\sqrt{5}}{5}$
 $= \sqrt{5}$

(e) $\sin \theta = 0.45$
 $\theta = 26^\circ 45'$

(f) $N 40^\circ W = 320^\circ$

(g) $3(6c)^2 - a + c$
 $= 3(6)^2 - 4 - 3$
 $= 101$

Q2
(a) $\cos 30^\circ + \sin 45^\circ$
 $= \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}}$
 $= \frac{\sqrt{6} + 2}{2\sqrt{2}}$

(b) $x^2 + 5x - 6 = (x+6)(x-1)$

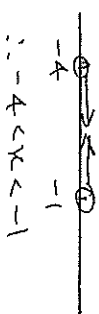
(c) $4x+7 = 3(x-2)$
 $4x+7 = 3x-6$
 $x = -13$

(d) (i) $3^{x-1} = 9^{2x}$
 $3^{x-1} = 3^2(2x)$

$x-1 = 4x$
 $-3x = 1$
 $x = -1/3$

(ii) $5 - 3x < 7$
 $-3x < 2$
 $x > -2/3$

(iii) $|2x+5| < 3$
 $2x+5 < 3$ OR $-(2x+5) < 3$
 $2x < -2$ $-2x-5 < 3$
 $x < -1$ $-2x < 8$
 $x > -4$



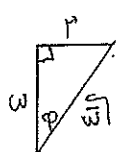
(e) $BC^2 = 550^2 + 400^2 - 2(550)(400) \cos 48^\circ$
 $= 168082.53$
 $BC = \sqrt{168082.53}$
 $BC = 410m$

Q3
(a) 0.23

(b) $x = 0.2323 \dots$
 $100x = 23.2323 \dots$
 $99x = 23$
 $x = \frac{23}{99}$

Q3 cont
(i) $\tan \theta = -\frac{3}{2}$, $\sin \theta > 0$
 $\cot \theta = -\frac{2}{3}$

(ii) $\sin \theta = \frac{2}{\sqrt{13}}$



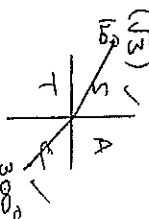
(c) $|x-1| = 4$

$x-1 = 4$ OR $-(x-1) = 4$
 $x = 5$ $-x+1 = 4$
 $-x = 3$
 $x = -3$

$\therefore x = 5, -3$

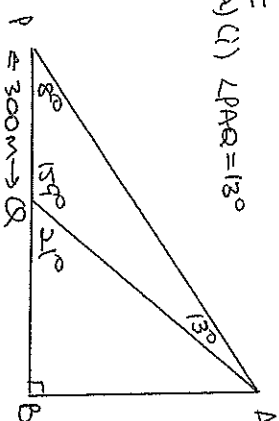
(d) $\sin \theta + \sqrt{3} \cos \theta = 0$, $0^\circ \leq \theta \leq 360^\circ$

$\sin \theta = -\sqrt{3} \cos \theta$
 $\tan \theta = -\sqrt{3}$
 $\tan \theta = -\sqrt{3}$
 $\theta = -\tan^{-1}(\sqrt{3})$
 $= -60^\circ$



$\therefore \theta = 120^\circ, 300^\circ$
 (a) $1 + \tan^2(90-\theta) = 1 + \cot^2 \theta$
 $= \operatorname{cosec}^2 \theta$

Q4
(a) (i) $\angle PAQ = 13^\circ$



(a) (ii) $AQ = \frac{300 \sin 80^\circ}{\sin 13^\circ}$

$AQ = \frac{300 \sin 80^\circ}{\sin 13^\circ}$
 $= 185.6m$ (1 d p)

(iii) $\sin 21^\circ = \frac{AB}{AQ}$

$\sin 21^\circ = \frac{AB}{185.6}$
 $AB = 185.6 \sin 21^\circ$
 $= 66.5m$ (1 d p)

(b) (i) $\frac{2x^2 - 3xy}{xy - y^2} \div \frac{4x - 6y}{2x^2 - 2xy}$
 $= \frac{x(2x-3y)}{y(x-y)} \times \frac{2x(x-y)}{2(2x-3y)}$
 $= \frac{x^2}{y}$

(ii) $\frac{a^3 + b^3}{a^2 - b^2} = \frac{(a+b)(a^2 - ab + b^2)}{(a+b)(a-b)}$
 $= \frac{a^2 - ab + b^2}{a-b}$

(iii) $\frac{2}{m^2 - 4} - \frac{1}{m^2 - 3m + 2}$
 $= \frac{2}{(m-2)(m+2)} - \frac{1}{(m-2)(m-1)}$
 $= \frac{2(m-2) - (m+2)}{(m-2)(m+2)(m-1)}$
 $= \frac{2m - 2 - m - 2}{(m-2)(m+2)(m-1)}$
 $= \frac{m - 4}{(m-2)(m+2)(m-1)}$

85

(a) (i) $x^2 + 15^2 = 30^2$

$x^2 = 675$

$x = \sqrt{675}$

$x = \frac{\sqrt{225} \times 3}{1}$
 $= 15\sqrt{3}$

(ii) $\sin \theta = \frac{15}{30}$

$= \frac{1}{2}$

$\therefore \theta = 30^\circ$

(b) (i) $\sin 240^\circ = -\frac{\sqrt{3}}{2}$

(ii) $\cos 30^\circ = \frac{\sqrt{3}}{2}$

(iii) $\csc 45^\circ = \frac{1}{\sin 45^\circ}$
 $= \frac{1}{\frac{1}{\sqrt{2}}}$
 $= \sqrt{2}$

(iv) $\cot 420^\circ = \cot 60^\circ$
 $= \frac{1}{\tan 60^\circ}$
 $= \frac{1}{\sqrt{3}}$

$= \frac{1}{\sqrt{3}}$

(c) $\frac{\sin \theta}{12} = \frac{\sin 30^\circ}{7}$

$\sin \theta = \frac{12 \sin 30^\circ}{7}$

$\theta = 59^\circ, 180 - 59^\circ$

$\therefore \theta = 59^\circ, 121^\circ$

(note that $121^\circ + 30^\circ = 151^\circ < 180^\circ$)

$\therefore 121^\circ$ is also valid

(d) $\frac{\sqrt{2}-1}{2\sqrt{2}-1} \times \frac{2\sqrt{2}+1}{2\sqrt{2}+1}$

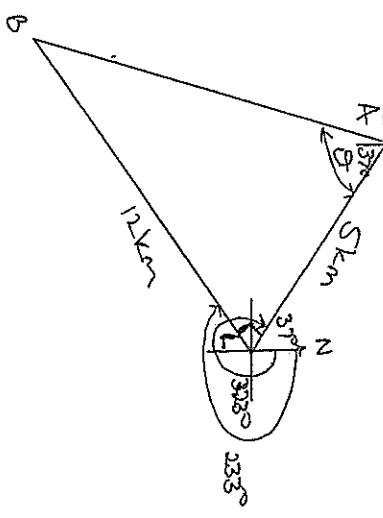
$= \frac{2\sqrt{2}-1}{4+2\sqrt{2}-2\sqrt{2}-1}$
 $= \frac{2\sqrt{2}-1}{3}$

$= \frac{3-\sqrt{2}}{7}$

$\therefore a = \frac{3}{7}, b = -\frac{1}{7}$

(e) $\sqrt{169a^2b^2} = 13a^3b$

(b) (i)



(ii) $AB^2 = 5^2 + 12^2$
 $= 169$
 $AB = 13 \text{ km}$

(iii) $\tan \theta = \frac{12}{13}$
 $\theta = \tan^{-1} (12/13)$
 $= 67^\circ 23'$

$\therefore$ bearing is $180^\circ + 30^\circ + 23'$
 $= 210^\circ 23'$

86 cont

(c) (i) $5^2 = 8^2 + 11^2 - 2(8)(11) \cos \angle ABC$

$25 = 185 - 176 \cos \angle ABC$
 $-160 = -176 \cos \angle ABC$

$\cos \angle ABC = \frac{10}{11}$

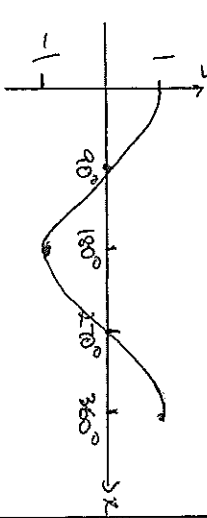
$\angle ABC = \cos^{-1} (\frac{10}{11})$
 $= 25^\circ$

(ii) $A = \frac{1}{2} \times 8 \times 11 \sin 25^\circ$
 $= 18.6 \text{ cm}^2 \text{ (1 d.p.)}$

(d) $2x^2 = x + 5$
 $2x^2 - x - 5 = 0$

$x = \frac{1 \pm \sqrt{(-1)^2 - 4(2)(-5)}}{2(2)}$
 $= \frac{1 \pm \sqrt{1+40}}{4}$
 $= \frac{1 \pm \sqrt{41}}{4}$

(e) $y = \cos \theta$



(b) $3x - 2y = 14 \dots \textcircled{1}$
 $xy = 12 \dots \textcircled{2}$

From $\textcircled{2}$ $y = \frac{12}{x} \dots \textcircled{3}$

sub $\textcircled{3}$ into $\textcircled{1}$
 $3x - 2(\frac{12}{x}) = 14$

$3x^2 - 24 = 14x$
 $3x^2 - 14x - 24 = 0$

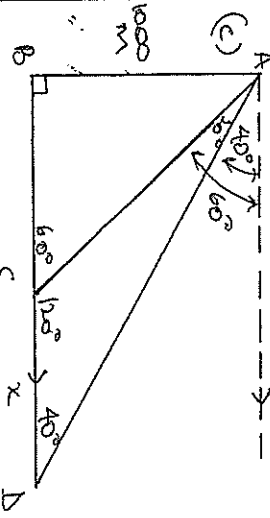
$(3x-18)(x+4) = 0$

$\frac{x}{3}(x-6)(3x+4) = 0$

$(x-6)(3x+4) = 0$
 $x = 6, -4/3$

when $x = 6, y = 2$
" $x = -4/3, y = -9$

$\therefore (6, 2) \text{ \& } (-\frac{4}{3}, -9)$



$\sin 60^\circ = \frac{1000}{AC}$
 $AC = \frac{1000}{\sin 60^\circ}$

$= 1154.7 \text{ m (1 d.p.)}$

$\frac{x}{\sin 20^\circ} = \frac{1154.7}{\sin 40^\circ}$

$x = \frac{1154.7 \sin 20^\circ}{\sin 40^\circ}$
 $= 614.4 \text{ m (1 d.p.)}$

87

(a) $x^2 - 10x - 1 = 0$
 $x^2 - 10x = 1$
 $x^2 - 10x + 25 = 1 + 25$
 $(x-5)^2 = 26$
 $x-5 = \pm \sqrt{26}$
 $x = 5 \pm \sqrt{26}$

88

Q7 cont

$$(d) (1 - \tan x)^2 + (1 + \tan x)^2 = 2 \sec^2 x$$

LHS

$$(1 - \tan x)^2 + (1 + \tan x)^2$$

$$= 1 - 2 \tan x + \tan^2 x + 1 + 2 \tan x + \tan^2 x$$

$$= 2 + 2 \tan^2 x$$

$$= 2(1 + \tan^2 x)$$

$$= 2 \sec^2 x$$

$$= \text{RHS.}$$

$$(e) x = 5 \sin \theta$$

$$\sin \theta = \frac{x}{5}$$

$$\sin^2 \theta = \left(\frac{x}{5}\right)^2$$

$$\text{Also, } y = 5 \cos \theta - 1$$

$$5 \cos \theta = y + 1$$

$$\cos \theta = \frac{y+1}{5}$$

$$\cos^2 \theta = \left(\frac{y+1}{5}\right)^2$$

$$\text{Now, } \sin^2 \theta + \cos^2 \theta = 1$$

$$\left(\frac{x}{5}\right)^2 + \left(\frac{y+1}{5}\right)^2 = 1$$

$$\frac{x^2 + (y+1)^2}{25} = 1$$